

Deep Unsupervised Learning using Nonequilibrium Thermodynamics

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I) Summary

This paper points at a solution to a thorny problem scientists have had to face for a while: modeling a complex set of data.

They usually have to choose between accuracy and flexibility: one can take any measurable function and transform it into a probability measure -being thus very flexible- however, the more accurate the function, the harder the normalization constant might be to compute.

Had one tried to generate data from distributions modeled with that method, there would have been a tradeoff between accuracy and flexibility.

In this work, the authors focus on two probability distributions, one we have samples from but not the expression of course, named q , the other one is p .

The goal is to go from q to p by the forward diffusion process and then to train the parameters of the reverse diffusion process so that the model can be used to generate data with q distribution from a sample of data which has p distribution.

The main idea of the diffusion model is the Markov chain, whether it is the forward or the backward process. At each timestep, the distribution p is updated, in this work mainly with a Gaussian diffusion process with μ_t mean and Σ_t covariance and a β_t diffusion rate.

The only difference between the two processes is that during the forward one, the parameters are known whereas they are trained during the backward process.

The authors show that the log-likelihood, used to measure how accurate an approximation of a distribution is, can be under bound by a quantity K . The maximization of K with respect to the parameters will give the steps of the reverse diffusion process that are the closest to the inverse of the trajectory that led in the first place from q to p .

This paper proposes a method inspired by statistical physics to transform a probability distribution into another through a diffusion-like process.

Viewing a probability distribution as the result of a diffusion process starting from another distribution allows to simplify the problem of generating data. Classically, a distribution would have had to be learned and then evaluated. With this work, we can simply learn the process of diffusion (only a few parameters at each timestep) and apply it directly to a sample of the starting distribution to get a generated piece of data.

II) Discussion of the main results:

One important consequence of that way of handling distributions is that multiplying the generated distribution with another becomes substantially easier.

Instead of brutally multiplying the two, we simply incorporate the multiplying distribution as a perturbation during each step of the diffusion process. The condition on the multiplying distribution is that it should change slowly over the process of diffusion. The authors ran the calculation for a Gaussian and a binomial diffusion process and give the perturbed Markov kernel.

One remarkable, while intuitive, property of the Gaussian diffusion process is that as long as the multiplying distribution $\mathbf{r}$ is smooth compared to the distribution $\mathbf{q}$, a Taylor expansion of the product of the distribution shows that the diffusion process is still Gaussian, with perturbation of its mean and covariance related to $\mathbf{r}$.

The choice of the diffusion rate is also addressed. With an analogy to thermodynamics, which states that the faster the process, the more entropy is created, the authors explain the importance of beta in the performance of the learning.

The authors are also able to give bounds for the conditional entropy of a single step of backward diffusion. Those bounds only depend on the forward trajectory. Even better, the upper bound for one step of the backward process (from t to $t-1$) is the entropy generated by the reverse step of the forward process (from $t-1$ to t). We can here better understand why the choice of beta was critical and why a beta too large, while speeding up the forward process, would make harder the learning of the backward process.

III) Limitations:

While the proposed method avoids the otherwise necessary tradeoff between flexibility and accuracy, a first limitation relies in the diffusion rate beta. A compromise is needed to get a fast enough forward diffusion process without making the learning too hard and therefore too long.

A second limitation of that method could be found when multiplying the generated distribution $\mathbf{q}$ by another one $\mathbf{r}$. As described before, $\mathbf{r}$ needs to be smooth and to vary slowly with the time steps to be considered as a perturbation.

Finally, this approach limits the generation of images to a class of images near the ones used for the training. Mixing classes of images (to be able to generate a cat or a dog on command) with that method requires to incorporate the classes during the learning process.

IV) Perspectives:

While the Gaussian and binomial distributions are judicious choices to begin with, they may lack the possibility of modeling a distribution with a tail. The next steps for that area of research could be to try for instance a Levy distribution to see what happens to the reverse Markov kernel.

I tried coding it in Python and, while the forward diffusion of course worked, the learning process involved being able to compute KL divergences which I lacked time to succeed in doing. I thought about brutally defining the KL divergences to Python and having it maximize them, to avoid analytical computations. However, I realized that the Levy distributions were not continuous let alone differentiable, and that using a gradient descent algorithm on such a function would be at least questionable, if not impossible.

This underlines that the advantage of the method of this paper relies in the possibility of transforming the problem into a learning problem with an objective function which can be computed and optimized with respect to its parameters.